Heat-engine efficiency

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate useful output and rejected energy for a heat engine.

SUCCESS CRITERIA - I can...

1. Use efficiency=work output/heat input.
2. Account for rejected heat.
3. Distinguish an ideal upper bound from actual performance.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “efficiency” and “cycle” to explain your model or result. Sentence starters are optional.

efficiency: useful output divided by input

cycle: a process that returns a system to its initial state

kelvin: the absolute temperature scale used in thermal ratios

01 Warm-up: bring back an earlier idea

For a fixed ideal-gas particle mass, average translational kinetic energy becomes 9 times as large.
Compare rms speed.

Word help: particle = a small constituent, such as an atom or molecule

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\eta = W_{out}/Q_{hot}$; $Q_{hot} = W_{out} + Q_{cold}$
$\eta_{max} = 1 - T_{cold}/T_{hot}$.
Cyclic heat engine; heat amounts are positive magnitudes. The reversible upper bound uses kelvin temperatures.

Worked reasoning: A heat engine takes 100 J from a hot source, does 25 J of work and rejects 75 J. Efficiency=25/100=25%.

Your turn: Draw three labeled energy arrows and check conservation.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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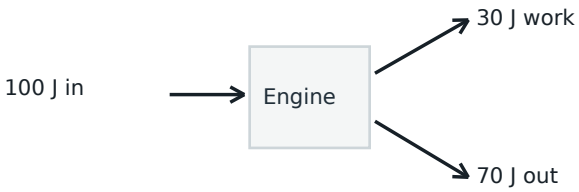
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

An engine receives 100 J, produces 30 J of work and rejects 70 J.
Check the energy balance and find efficiency.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

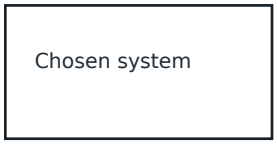
04 Apply the physics

A model heat engine operates between reservoirs at 600 K and 300 K. It receives 800 J per cycle.
Find the ideal maximum efficiency and work. Its measured work is 240 J; find actual efficiency and complete the actual energy account.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why should a 95% efficient electric motor not be confused with a 95% heat engine converting heat from one source?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

At fixed speed, mass becomes 4 times as large while energy-coupling fraction becomes 1/6 as large.
Compare coupled energy.
Show the energy changes and transfers; label units.

Sentence starter: Both ... . They differ in ... because ... .



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A heat engine has Q_{hot}=1500 J and Q_{cold}=1050 J.
Find work and efficiency.
Label energy entering, leaving or stored on the system model.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

An ideal upper bound is $\eta=1-T_c/T_h$ with kelvin temperatures. For T_h=600 K and T_c=300 K, compare that bound with an engine producing 400 J work from 1000 J input.
Label energy entering, leaving or stored on the system model.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3